Mengchen Fan

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Summary

- My research develops AI methodologies for learning from complex and heterogeneous data. I focus on efficient and reliable models, including model compression and adaptation, multimodal data fusion, longitudinal-data reasoning, and distributed learning. Medical imaging is a primary application area for my research.
- **Research Interests:** Medical AI, Distributed Learning, and Federated Learning.
- **Teaching Interests:** Machine Learning, Deep Learning, Programming, and Algorithms.

Education

- **University of Alabama at Birmingham** — *Aug 2022 – Present*
  Ph.D. in Computer Science, GPA: 4.0/4.0; Expected May 2027 — Birmingham, AL, United States
- **Syracuse University** — *Sept 2017 – May 2019*
  M.S. in Computer Engineering, GPA: 3.867/4.0 — Syracuse, NY, United States
- **North China Electric Power University** — *Sept 2012 – May 2016*
  B.S. in Electrical Engineering, GPA: 3.47/4.0 — China

Research Experience

- **University of Alabama at Birmingham** — *Aug 2022 – Present*
  Doctoral Researcher in Computer Science — Birmingham, AL, United States
  - Developed region-aware distillation methods that produced lightweight 3D medical-image segmentation models.
  - Developing longitudinal MRI/CT reasoning benchmarks and agent-based methods for medical image understanding and reasoning.
  - Developed federated and decentralized learning methods for heterogeneous clients through personalized training and representative knowledge sharing.
  - **Outcomes:** Twelve peer-reviewed publications, including nine first or co-first author papers. The MICCAI 2026 paper received an early-accept decision (top 9% of 4,601 submissions) and was selected for a Spotlight presentation. My work on longitudinal brain MRI reasoning was also accepted to NeurIPS 2026. Two papers were accepted to WACV 2027.
- **Pulin Che Laboratory, UAB Heersink School of Medicine** — *May – Aug 2026*
  Research Intern — Birmingham, AL, United States
  - Developed deep learning pipelines for ImageStreamX imaging flow cytometry to quantify pathway-specific Golgi morphology.
  - Developed interpretable classifiers and validation methods for inflammatory activation, signaling-pathway identification, and generalization across species and unseen compounds.
  - **Outcomes:** Three submitted manuscripts on Golgi morphology analysis, signaling-pathway identification, and drug repurposing.

Publications

J–Journal, C–Conference, A–Abstract, P–Patent

†Equal contribution

[C.1] **Mengchen Fan**†, Qizhen Lan†, Xi Xiao, Lin Zhao, Liang He, Tianyang Wang, Xiaoqian Jiang, Pulin Che, and Baocheng Geng. *MedSTP: Preserving Subspace and Transition Evidence for Medical VQA Token Pruning*. **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027.

[C.2] Baraa Bilbeisi†, **Mengchen Fan**†, Baocheng Geng, and Qing Tian. *CRAD: Class-wise Reliability-Aware Distillation for Decentralized Heterogeneous Federated Learning*. **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)**, 2027.

[C.3] Qizhen Lan†, **Mengchen Fan**†, Hang Zhang, Jingwei Duan, Moule Lin, Jialin Chen, Baocheng Geng, and Xiaoqian Jiang. *BrainTRACE: Tracing Longitudinal, Multimodal, and Volumetric Evidence in Brain MRI Clinical Reasoning*. **Conference on Neural Information Processing Systems (NeurIPS)**, 2026.

[C.4] **Mengchen Fan**, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang, and Qizhen Lan. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)**, 2026, **Early Accept (top 9%); Spotlight**.

[C.5] **Mengchen Fan**, Fei Zhao, Chengcui Zhang, Shuya Feng, and Baocheng Geng. *On Optimal Modality Compression for Bandwidth Constrained Multimodal Fusion*. **IEEE International Conference on Information Reuse and Integration for Data Science (IRI)**, 2026.

[C.6] **Mengchen Fan**, Tianyun Zhang, and Baocheng Geng. *A Unified Framework for the Convergence and Weight Pruning in Federated Learning*. **ACM Multimedia Asia**, 2025.

[C.7] **Mengchen Fan**, Keren Li, Tianyun Zhang, Qing Tian, and Baocheng Geng. *PFedDST: Personalized Federated Learning with Decentralized Selection Training*. **IEEE International Joint Conference on Neural Networks (IJCNN)**, 2025.

[J.1] **Mengchen Fan**, Tianyun Zhang, Xiaolong Ma, Jiacheng Guo, Zheng Zhan, Shanglin Zhou, Minghai Qin, Fei Sun, Yen-Kuang Chen, Caiwen Ding, Makan Fardad, and Yanzhi Wang. *A Unified DNN Weight Compression Framework Using Reweighted Optimization Methods*. **Intelligent Systems with Applications**, 2025, vol. 27.

[J.2] Joseph Casey, Qianjiao Chen, **Mengchen Fan**, Baocheng Geng, Roman Shterenberg, Zhong Chen, and Keren Li. *Distributed Collaborative Learning with Representative Knowledge Sharing*. **Mathematics**, 2025, vol. 13, no. 6, 1004.

[A.1] E. Chesnokova, D. J. Barrett, B. Geng, **Mengchen Fan**, T. Kulkarni, and S. Rothenberg. *An AI Physical Exam Can Predict Lung Function*. **American Thoracic Society (ATS) International Conference**, 2025, Poster Discussion Abstract, A5307.

[C.8] **Mengchen Fan**, Baocheng Geng, Keren Li, Xueqian Wang, and Pramod K. Varshney. *Interpretable Data Fusion for Distributed Learning: A Representative Approach via Gradient Matching*. **International Conference on Information Fusion (FUSION)**, 2024, pp. 1–8.

[J.3] Yusen Zhai, **Mengchen Fan**, Baocheng Geng, Xue Du, Scott Snyder, and Larrell Wilkinson. *Impact of Phased COVID-19 Vaccine Rollout on Anxiety and Depression among the U.S. Adult Population, January 2019–February 2023: A Population-Based Interrupted Time Series Analysis*. **The Lancet Regional Health – Americas**, 2024.

[P.1] Jinju Qian, Zhichao Mo, Zhangquan Rao, Weibing Yue, Ke Wang, **Mengchen Fan**, Xiangyang Peng, Xiaofeng Pang, Lin Yi, and Rui Wang. *Routing Inspection Path Planning Method and Device for Routing Inspection Robot*. **Chinese Patent CN106156888B**, 2020.

Manuscripts Submitted and Under Review

[†]Equal contribution

[U.1] Qizhen Lan, Xi Xiao, Xiangchen Guan, **Mengchen Fan**, Moule Lin, Jung Im Choi, and Lijing Zhu. *Trust Is Not Enough: Influence Calibration for On-Policy Self-Distillation in Agentic RL*. **AAAI Conference on Artificial Intelligence**, 2027, under review.

[U.2] **Mengchen Fan**[†], Xu Zhang[†], Baocheng Geng, and Pulin Che. *SIGMA: SIgnaling via Golgi Morphology Analysis for Pathway-State Classification*. **Communications Biology**, 2026, submitted.

[U.3] Xu Zhang[†], **Mengchen Fan**[†], Kim Seokoon, Qing Zhao, Brant Wagener, Baocheng Geng, and Pulin Che. *Machine Learning Classifies Subcellular Golgi Morphology from Imaging Flow Cytometry during Inflammatory Activation: An Exploratory Study*. **npj Imaging**, 2026, submitted.

[U.4] Xu Zhang, **Mengchen Fan**, Xinnan Li, and Pulin Che. *Drug Repurposing from Single-Cell IPF Transcriptomes Identifies the HDAC Inhibitor Belinostat as an Antifibrotic Candidate*. **Respiratory Research**, 2026, submitted.

[U.5] **Mengchen Fan**, Roman Shterenberg, and Keren Li. *A Statistical Framework for Measuring Data Heterogeneity in Distributed Learning*. **The Annals of Applied Statistics**, 2025, under review.

Teaching and Mentoring Experience

University of Alabama at Birmingham

Spring 2026	**Natural Language Processing CS 662/762**, Teaching Assistant, Guest Instructor
Spring 2026	**Deep Learning CS 665/765**, Teaching Assistant, Guest Instructor
Fall 2025	**Computer Security CS 436/636/736**, Teaching Assistant
Summer 2025	**Mobile Application Development CS 422/522**, Teaching Assistant
Spring 2025	**Deep Learning CS 665/765**, Teaching Assistant, Guest Instructor
Fall 2024	**Foundations of Data Science CS 685**, Teaching Assistant
Summer 2024	**Mobile Application Development CS 422/522**, Teaching Assistant
Spring 2024	**Deep Learning CS 665/765**, Teaching Assistant, Guest Instructor
Fall 2023	**Machine Learning CS 667/767**, Teaching Assistant
Summer 2023	**Machine Learning CS 667/767**, Teaching Assistant, Guest Instructor

2025 – 2026	**Student Mentoring: Baraa Bilbeisi**, Federated learning, and knowledge distillation
2025 – 2026	**Student Mentoring: Yifan Ying**, Federated learning, and token merging

Grant and Proposal Contributions

- **UAB REINVENT Seed Funding Program** ($50,000), 2025. *Machine Learning-Based Early Detection of Acute Respiratory Distress Syndrome* (funded). Proposal Contributor; developed the machine-learning methodology and contributed to proposal writing. PI: Pulin Che.

Academic Service

- **Conference Reviewer:** AAAI (2026, 2027); MICCAI (2026); IJCNN (2025, 2027); ICME (2025); BioKDD (2025, External Reviewer).
- **Journal Reviewer:** IEEE/ACM Transactions on Computational Biology and Bioinformatics, 2024–2025; BMJ Open Ophthalmology, 2025; Journal of Clinical Pharmacy and Therapeutics, 2025; Frontiers in Genetics, 2025; Frontiers in Nutrition, 2025; Discover Oncology, 2025; Journal of Visualized Experiments (JoVE), 2026.

Invited Talk

- **Distributed Model Pruning and Data Fusion** — *June 10, 2024*
  2024 WNAR/IMS/Graybill Annual Meeting, Colorado State University — Fort Collins, CO, United States

Professional Experience

- **Citi** — *Dec 2020 – Aug 2022*
  Java Full Stack Developer — Shanghai, China

- Developed and maintained a full-stack SMA platform for Basel IV operational-risk calculations using Java, Spring, Angular, and Oracle; mentored new team members.

Technical Skills

Programming and Tools: Python, C++, Java, SQL, JavaScript; PyTorch, Spring, Angular, and Oracle.